

WATER HARVESTING

Rainwater harvesting, in its **broadest sense**, is a **technology used for collecting and storing rainwater for human use from rooftops**, land surfaces or rock catchments using simple techniques such as jars and pots as well as engineered techniques. Rainwater harvesting has been practiced for more than 4,000 years, owing to the temporal and spatial variability of rainfall. It is an **important water source in many areas with significant rainfall but lacking any kind of conventional, centralised supply system**.

Benefits Of Rain Water Harvesting for The Community

- It is also a good option in areas **where good quality fresh surface water or ground water is lacking**.
- Water harvesting enables **efficient collection and storage of rainwater**, makes it accessible and substitute for poor quality water.
- **Improvement in the quality of ground water**
- Rise in the **water levels in wells** and bore wells that are drying up,
- **Mitigation of the effects of drought** and attainment of drought proofing,
- **An ideal solution in areas having inadequate water resources**,
- Reduction in the **soil erosion as the surface runoff** is reduced,
- **Decrease in the choking of storm water** drains and flooding of roads and
- **Saving of energy** to lift ground water.

Types of Water Harvesting

1. Rainwater Harvesting:

Rainwater harvesting is **defined as the method for inducing, collecting, storing and conserving local surface runoff for agriculture** in **arid and semi-arid** regions. Three types of water harvesting are covered by rainwater harvesting.

2. Flood Water Harvesting:

Flood water harvesting can be **defined as the collection and storage of creek flow for irrigation use**. Flood water harvesting, also known as '**large catchment water harvesting**' or '**Spate Irrigation**'.

3. Groundwater Harvesting:

Groundwater dams like ‘**Subsurface Dams**’ and ‘**Sand Storage Dams**’ are other fine examples of groundwater harvesting. **They obstruct the flow of ephemeral streams** in a river bed; the **water is stored in the sediment below ground surface** and can be used for aquifer recharge.

Mechanism Of Rain Water Harvesting

1. Roof Top Harvesting

This system is **useful mainly for drinking water** purposes

- In this system, **rain water falling on roofs of houses and other buildings** is collected through a system of pipes and semi-circular channels of galvanized iron or PVC **and is stored in tanks suitably located on the ground or underground.**



Figure 1- Roof Top Rain Water Harvesting

- This system can be seen in **the northeastern states of Arunachal Pradesh, Assam, Meghalaya, Manipur and Nagaland.** This is also in use in **Bikaner, Jaisalmer and Jodhpur districts of Rajasthan.**

2. Tanka/ Kund/ Kundi

In the **desert and arid areas of Rajasthan** people build **unique underground structures of various shapes and sizes to collect rain water for drinking purposes.** These are built both for individual households as well as for village communities using locally available

materials. These simple **traditional water harvesting structures** are useful even during years of below-normal rainfall. These structures called **Tankas, Kunds or Kundis** are constructed in a variety of places like court yards, in front of houses and temples, in open agricultural fields, barren lands etc



Figure 2- Tanka

3. Ponds/ Tanks



This is by far the most commonly used method **to collect and store rain water in dug ponds or tanks.**

Figure 3 – Ponds (Water Harvesting Structure)

Table – Different Name Of Pond Structure In Different Region

Water harvesting structure (Pond)	Region
Nagaland	Zabo
Gujarat	Kunda (Sacred Ponds), jheel
Maharashtra	Maharashtra – Bandharas (Bunds across small streams)
Assam	Dong
Uttar Pradesh	Talab, Johad or Pokhar
Rajasthan	Johad or Nadi
Ladakh	Zing
Bihar	Pyne ahar

4. Ground Water Harvesting

In hilly areas of **Uttaranchal**, the people **harvest ground water** by **making stonewall across ground water streams**. These are called **Naula or Hauzi**, in **Rajasthan** these are called **Beris**, **Gujarat** shallow wells dug in depressions to tap ground water are called **Virdas**



Figure 5 – Beris Of Rajasthan

5. Khadin

Khadin is a system of **storing rain water in an agricultural field by building a “U” shaped earthen bund at the lower end of the field**. A drainage pipe is provided in the embankment to evacuate surplus water. The practice is very common in the arid and **semi-arid areas of Jaisalmer and Barmer districts of Rajasthan**.

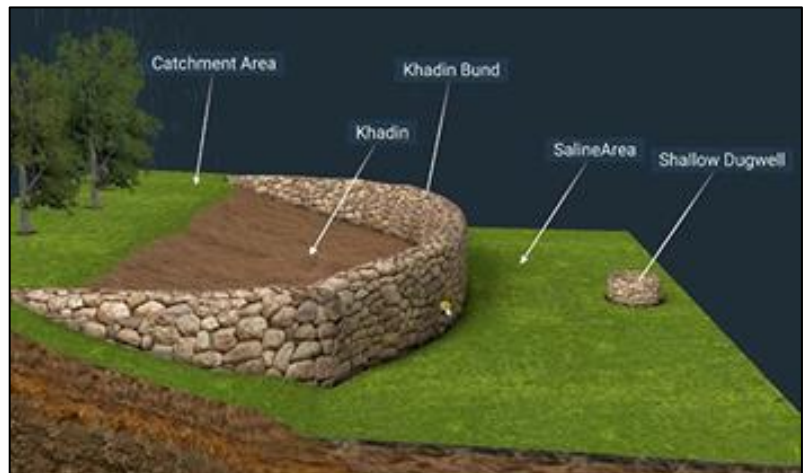


Figure 6 – Khadin In Rajasthan

A similar system called **Haveli** is used in some parts of **Madhya Pradesh**.

6. Hill Slope Collection

In this system, which is in **vogue in many hilly areas with good rainfall**, **lined channels are built across the hill slopes to intercept rain water**. These channels convey water for irrigating terraced agricultural fields. The water is also used

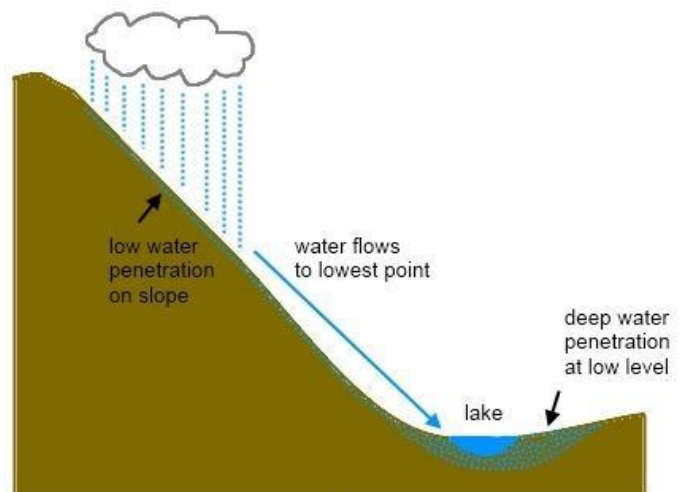


Figure 7 – Hill Slope Water Collection

to fill small ponds for domestic use and the cattle. **These practices can be seen in Uttaranchal, Himachal Pradesh, Meghalaya, Arunachal Pradesh etc**

7. Spring Water Harvesting

Spring water **is a highly desirable source of community water supply**. Since the water emerges at the ground surface through cracks and loose joints in rocks under internal pressure of the ground water system, no pumping is required. In **the Lahaul and Spiti areas of Himachal Pradesh, water from hill streams are diverted through small excavated channels, called Kuls, for domestic use and irrigation. In Jammu region they pronounce it as Kuhals.**



Figure 8 – Kuls in Himachal Pradesh

This practice can also be **seen in Arunachal Pradesh, Meghalaya, Nagaland, Manipur, Sikkim and Darjeeling** area of West Bengal.

Modern Structures

The structures, which are **commonly built for surface storage and/ or ground water recharge** are

1. Check Dams-

These are **concrete or masonry structures** built across small streams for **surface storage and incidental benefit of ground water** recharge. These are usually built by **the State Government agencies** like the departments of Irrigation/ Water Resources, Agriculture and Forests.



Figure 9 – Check Dam

2. Percolation Tanks

These are built mainly to **impound monsoon runoff over a large area** to augment ground water recharge. Moderate **to high porosity of soil and/ or underlying rocky** strata is the main criteria for the choice of percolation tanks.



Figure 10 – Percolation Tank

3. Sub-Surface Dykes

These are **impermeable walls or barriers in masonry, concrete and/ or clay built below the bed level across natural streams** to arrest sub-surface flow of water to improve the yield of existing wells and hand pumps in the upstream.

Some Indirect Methods Have Also Been Developed

1. Contour Bunding:

These are small earthen **bunds built horizontally in parallel rows across the hill slope**. These help in augmenting soil moisture and prevent erosion of topsoil.

Figure 11 – Contour Bunding



Figure 12 – Gully Plugging

2. Gully Plugging:

These are soil and water retaining structures built **across gullies in hilly areas**. These are built with locally available **materials like stone boulders, earth, brushwood** etc.

Significance Of Rain Water Harvesting:

- Ensuring **food security**
- Modern **hybrid crops & fertilizer depend on continuous supply of irrigation** water.
- More than **60% of net sown area in India is rainfed and to address the rising cases of drought in the dry belt of India and ensure food security** RWH is essential.
- More **than 3/4th of the precipitation In India occurs during 4 months of monsoon** season and significant part of which is lost in runoff and evaporation, this can be **controlled by rain water harvesting**.
- **Rapid urbanization and Industrial development demand** huge water hence severe water crunch can be addressed by water harvesting.
- In dry **and hilly areas women face hardship in fetching water** and in such remote areas RWH can provide a decentralized source of water which would **reduce their hardships**.

Way Forward

- There is a need to enhance efforts to **reduce dependency on groundwater** with a focus on
- Enhancing **water security**
- **Rejuvenation and recharge** of natural and constructed water bodies.
- **Increased reuse of treated wastewater** through local sewage treatment plants.
- **Build well-designed rainwater harvesting systems** across residential and commercial areas.
- These efforts must be **facilitated by cooperation of Central and State governments**
- There is a need for **stringent policy initiatives** for long-term and sustainable solutions.
- An **integrated water management plan** must be brought up by local bodies and implemented at the local level.